

EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol for AI Agents over Foundation-Model Embeddings

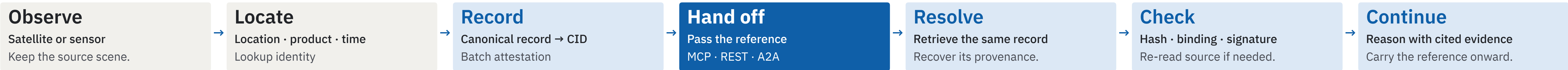
An Earth observation that survives an agent handoff.

EMEM gives a sampled or derived observation a reusable reference. The next agent recovers its value, location, acquisition time and processing recipe, then checks the evidence.

A compact citation keeps the observation traceable across agents and time. Encode in orbit, decode in AI's reasoning with emem.

Jaya Kumari · Avijeet Singh · Vortex AI · avijeet@vortex.ai
Agentic AI for Earth Observation · BIFOLD and ESA D-lab · Berlin · Poster Session 1 · 19 Oct 2026 · emem.dev · github.com/Vortex-AI/emem (Apache-2.0)

From observation to the next agent



1 What survives an agent handoff?

The same observation, carried through receiver checks.

Agent A cites a source read. A relay changes the evidence. Agent B acts on it or checks the reference.

